

SST Derivation Note: Bell-like Phase Witness for Swirl-String Channels

1 Purpose

This note translates the logic of momentum-entangled Bell-correlation experiments into a Swirl-String Theory (SST) framework. The goal is not to claim that SST already reproduces quantum Bell nonlocality. The goal is narrower:

Can an SST medium with paired counter-propagating or topologically conjugate excitation channels produce a phase-sensitive correlation witness of Bell-like sinusoidal form under controlled recombination?

This is a well-posed operational question even before any ontological claim is made.

2 Orthodox summary

In the atom-interferometric experiment, pair-correlated massive particles are generated in opposite momentum modes by collisions of ultracold condensates. In the low-occupancy limit, the relevant four-mode sector is approximated by a Bell-like state. A matter-wave Rarity–Tapster interferometer then mixes the two branches and yields joint probabilities of the form

$$P_{pp'} = \frac{1}{2} \sin^2\left(\frac{\Phi}{2}\right), \quad P_{pq'} = \frac{1}{2} \cos^2\left(\frac{\Phi}{2}\right), \quad (1)$$

with analogous relations for the other pairings. The experimentally constructed Bell-correlation function is

$$E(\Phi) = \frac{P_{pp'} + P_{qq'} - P_{pq'} - P_{qp'}}{P_{pp'} + P_{qq'} + P_{pq'} + P_{qp'}} \approx -A \cos(\Phi + \delta), \quad (2)$$

where A is the visibility and δ is an offset phase. The measured amplitude is large enough to witness Bell correlations and steering-type nonlocality in a motional degree of freedom of massive atoms. A steering witness of the form

$$C(\Phi, \Phi + \pi) = |E(\Phi) - E(\Phi + \pi)| \quad (3)$$

is then tested against a threshold. The key observable fact is not merely pair production, but phase-controlled oscillatory correlations after recombination.

3 SST mapping

We now define the SST translation.

3.1 Operational map

The orthodox objects and their SST analogs are:

Orthodox object	SST analog
Opposite momentum modes	paired swirl-string propagation channels
Two-mode squeezed / Bell-like sector	paired channel occupancy state
Bragg mirror / beamsplitter pulses	controlled coupling junctions between channels
Interferometric phase Φ	accumulated swirl phase / delay phase
Coincidence probabilities	paired detection rates at output ports
Bell correlation $E(\Phi)$	Bell-like SST phase witness

3.2 SST interpretation

In SST, the basic hypothesis is that a localized source event can launch a pair of coherent excitations into two alternative channel pairs,

$$(K_a, K'_a) \quad \text{or} \quad (K_b, K'_b),$$

where K_i may denote:

- (a) geometric channels in a recombination network,
- (b) counter-propagating swirl-string branches,
- (c) topologically conjugate mode sectors,
- (d) or two linked transport pathways through the foliation field.

The physically meaningful claim is therefore:

If SST supports pair creation into two coherent alternatives, and if the two alternatives can be recombined with controllable phase delay, then a sinusoidal correlation witness should emerge at the output.

4 Minimal SST equations

4.1 Paired channel state

As a minimal effective model, take the reduced paired-channel state

$$|\Psi_{\text{SST}}\rangle = \frac{1}{\sqrt{2}} \left(|1\rangle_{K_a} |1\rangle_{K'_a} |0\rangle_{K_b} |0\rangle_{K'_b} + e^{i\chi} |0\rangle_{K_a} |0\rangle_{K'_a} |1\rangle_{K_b} |1\rangle_{K'_b} \right), \quad (4)$$

where χ is the source-imprinted relative phase.

This is a *model ansatz*, not yet derived from Canon. Its purpose is to define the simplest paired superselection sector that could support Bell-like phase readout.

4.2 Channel phase accumulation

Let the two output arms L and R accumulate phases

$$\phi_L = \int_L (k_{\text{eff}} d\ell - \omega_{\text{eff}} dt) + \phi_L^{(\text{junc})}, \quad \phi_R = \int_R (k_{\text{eff}} d\ell - \omega_{\text{eff}} dt) + \phi_R^{(\text{junc})}, \quad (5)$$

with total phase

$$\Phi_{\text{SST}} = \phi_L + \phi_R + \chi. \quad (6)$$

In an SST medium, the effective local frequency may depend on the swirl-clock sector,

$$\omega_{\text{eff}}(\mathbf{x}) = \omega_0 \mathcal{F}(S_t^\odot(\mathbf{x}), \rho_f(\mathbf{x}), \rho_E(\mathbf{x}), \Omega(\mathbf{x})), \quad (7)$$

where Ω denotes local vorticity magnitude or an SST-compatible circulation-density proxy.

A minimal first-order closure is

$$\omega_{\text{eff}}(\mathbf{x}) \approx \omega_0 \left[1 + \alpha_S (1 - S_t^\odot(\mathbf{x})) + \beta_S \frac{\rho_E(\mathbf{x})}{\rho_f(\mathbf{x}) c^2} \right], \quad (8)$$

with dimensionless coefficients α_S, β_S . This is phenomenological.

4.3 Output probabilities

Assume the recombination junction acts as a balanced linear coupler. Then the coincidence probabilities have the same harmonic form as the orthodox case:

$$P_{aa'}(\Phi_{\text{SST}}) = P_{bb'}(\Phi_{\text{SST}}) = \frac{\eta}{2} \sin^2 \left(\frac{\Phi_{\text{SST}}}{2} \right), \quad (9)$$

$$P_{ab'}(\Phi_{\text{SST}}) = P_{ba'}(\Phi_{\text{SST}}) = \frac{\eta}{2} \cos^2 \left(\frac{\Phi_{\text{SST}}}{2} \right), \quad (10)$$

where $0 \leq \eta \leq 1$ is the effective pair-transfer efficiency.

4.4 Bell-like SST phase witness

Define

$$E_{\text{SST}}(\Phi) = \frac{P_{aa'} + P_{bb'} - P_{ab'} - P_{ba'}}{P_{aa'} + P_{bb'} + P_{ab'} + P_{ba'}}. \quad (11)$$

Substituting Eqs. (9)–(10) gives

$$E_{\text{SST}}(\Phi) = -\cos \Phi_{\text{SST}}. \quad (12)$$

Allowing for mode mismatch, dephasing, finite bandwidth, or unequal couplers yields

$$E_{\text{SST}}(\Phi) = -A_{\text{SST}} \cos(\Phi_{\text{SST}} + \delta_{\text{SST}}), \quad 0 \leq A_{\text{SST}} \leq 1. \quad (13)$$

This is the proposed *Bell-like phase witness for swirl-string channels*.

5 SST-specific source constraint

A more SST-native formulation replaces the abstract source phase χ by a local pair-creation law. The source region should conserve:

$$\Gamma_{\text{in}} = \Gamma_{\text{out},1} + \Gamma_{\text{out},2}, \quad (14)$$

$$\mathbf{K}_{\text{in}} \approx \mathbf{K}_1 + \mathbf{K}_2, \quad (15)$$

$$\omega_{\text{src}} \approx \omega_1 + \omega_2, \quad (16)$$

where Γ is circulation and $\mathbf{K}$ is the effective channel wavevector or transport momentum. A local source satisfying Eqs. (14)–(16) is the natural SST analog of opposite-momentum pair production.

6 Observables

The minimal observables are:

6.1 Visibility

$$A_{\text{SST}} = \frac{E_{\text{max}} - E_{\text{min}}}{2}. \quad (17)$$

A nonzero A_{SST} indicates preserved pair coherence through the SST channel network.

6.2 Clock-field phase shift

If one arm traverses a region of modified swirl clock S_t° , then

$$\Delta\Phi_{\text{clock}} = - \int \Delta\omega_{\text{eff}}(t) dt. \quad (18)$$

Thus, SST predicts that branch-dependent clock structure should appear as a measurable phase offset δ_{SST} .

6.3 Contrast loss from inhomogeneity

If the phase fluctuates over the detection window with variance σ_Φ^2 , then

$$A_{\text{SST}} \approx A_0 e^{-\sigma_\Phi^2/2}. \quad (19)$$

This gives a direct quantitative reason to restrict to narrow mode windows, exactly as in the atomic experiment.

6.4 Topological dependence

If the pair channels correspond to different knot or link sectors, one may test

$$A_{\text{SST}} = A_{\text{SST}}(K_a, K_b), \quad \delta_{\text{SST}} = \delta_{\text{SST}}(K_a, K_b), \quad (20)$$

which would probe whether topological complexity affects coherence preservation.

7 Falsifiers

This proposal is useful only if it can fail.

1. **No sinusoidal law.** If recombination does not produce

$$E_{\text{SST}}(\Phi) \approx -A_{\text{SST}} \cos(\Phi + \delta_{\text{SST}}),$$

then the Bell-like phase witness fails.

2. **No pair-coincident enhancement.** If source events do not generate statistically enhanced paired output rates relative to uncorrelated controls, then the paired-channel ansatz in Eq. (4) is not appropriate.
3. **No controlled phase shift.** If deliberately changing one arm's delay or local swirl environment does not shift δ_{SST} , then the interferometric mapping is wrong.
4. **No window dependence.** If narrowing the selected mode sector does not improve visibility, then the coherence-loss model in Eq. (19) is likely incorrect.
5. **No topology dependence.** If channel topology is claimed to matter, but A_{SST} and δ_{SST} remain invariant under channel substitution, then that claim should be rejected.

8 Status

- **Orthodox status:** established for ultracold-atom momentum-entanglement experiments.
- **SST status of the witness form:** plausible operational translation.
- **SST status of the source ansatz Eq. (4):** conjectural.
- **SST status of the clock-sensitive phase law Eq. (8):** phenomenological and not yet canonically derived.
- **SST status of Bell nonlocality claim:** not established.

9 Numerical SST scales

Using the canonical constants

$$\mathbf{v}_\mathcal{G} = 1.09384563 \times 10^6 \text{ m s}^{-1}, \quad r_c = 1.40897017 \times 10^{-15} \text{ m}, \quad \rho_f = 7.0 \times 10^{-7} \text{ kg m}^{-3},$$

the characteristic circulation scale is

$$\Gamma_0 = 2\pi r_c \mathbf{v}_\mathcal{G}. \quad (21)$$

Numerically,

$$\Gamma_0 = 2\pi(1.40897017 \times 10^{-15})(1.09384563 \times 10^6) \approx 9.6847 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}. \quad (22)$$

A corresponding characteristic angular frequency at the core scale is

$$\omega_c \sim \frac{\mathbf{v}_\mathcal{G}}{r_c} \approx \frac{1.09384563 \times 10^6}{1.40897017 \times 10^{-15}} \approx 7.7649 \times 10^{20} \text{ s}^{-1}. \quad (23)$$

Hence a path-delay difference Δt would induce a raw phase shift

$$\Delta\Phi \sim \omega_c \Delta t. \quad (24)$$

For example,

$$\Delta t = 10^{-21} \text{ s} \implies \Delta\Phi \sim 0.776. \quad (25)$$

This indicates that an SST phase witness would be extremely sensitive to even tiny effective delay differences if the relevant carrier frequency is indeed set by a core-scale mode. That sensitivity is useful, but also means averaging and dephasing are serious confounders.

10 Minimal experiment or simulation

A minimal SST testbed needs only four ingredients:

- (i) a source that launches paired excitations into two alternative channel pairs;
- (ii) two controllable couplers acting as mirror and beamsplitter analogs;
- (iii) a tunable delay or local swirl-clock perturbation in one arm;
- (iv) coincidence-counting or paired-output statistics.

Even a classical wave-network simulation is enough to test whether the witness law Eq. (13) appears. One need not begin with a full quantum claim.

11 Conclusion

The most important lesson from the orthodox experiment is not “Bell violation” by itself, but the operational chain:

pair source $\rightarrow$ branch coherence $\rightarrow$ phase accumulation $\rightarrow$ recombination $\rightarrow$ oscillatory correlation witness

SST can adopt exactly this chain. The proposed quantity

$$E_{\text{SST}}(\Phi) = -A_{\text{SST}} \cos(\Phi + \delta_{\text{SST}})$$

is therefore a clean first target: measurable, falsifiable, and naturally sensitive to channel delay, medium inhomogeneity, and possibly topological class.

Analogy for a 10-year-old. Imagine making two matching ripples in water and sending them through two different maze paths. If the paths keep the ripples coordinated, then when they meet again they make a repeating strong-weak pattern. That repeating pattern is what this note tries to measure in SST.

References

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